

22. (2)

$$2 \times (3 \times 4 \times 5) + 2 \times (2 \times 3 \times 4 \times 5) + 4 \times (2 \times 3 \times 5)$$

$$\pi 2 \times 3 \times 4 \times 5$$

$$= \sum f_m(s_k) \times \frac{f_n(s_k)}{h}$$

$$\div \sum_{i=1}^n f_n(s_i)$$

$$\frac{1}{h} \sum_k \left(f_m(s_k) \pi_{i \neq k} f_n(s_i) \right)$$

$$\frac{3}{5} \left(\frac{1}{h} \sum_j f_m(s_j) \pi_{i \neq j} f_n(s_i) \right) + \frac{2}{5} \left(\frac{1}{h} \sum_{i=1}^n \pi_{i \neq i} f_n(s_i) \right)$$

$$= \frac{1}{h} \sum_k \left(f_m(s_k) / f_n(s_k) \right)$$

$$f_m(s_k) \pi_{i \neq k} f_n(s_i)$$

$$\frac{3}{5} \left(\frac{1}{h} \sum_j f_m(s_j) / f_n(s_j) \right) + \frac{2}{5} n$$

$$\frac{1}{h} \sum \frac{f_m}{f_n} = \frac{1}{h} \sum R_k = \Lambda_{org}$$

$$= \left(\frac{1}{h} \sum_k \frac{f_m(s_k)}{f_n(s_k)} \right) = \frac{1}{h} \sum_{k=1}^n R_k$$

$$= \frac{\Lambda_{org}}{\pi(\text{confuse}) \times \Lambda_{org} + (1 - \pi)}$$

$$\left(\frac{3}{5} \times \frac{1}{h} \sum_j \frac{f_m(s_j)}{f_n(s_j)} + \frac{2}{5} \right) = \frac{3}{5} \frac{1}{h} \sum_j R_k + \frac{2}{5}$$

Λ_{new}

(1)

$$P(f_m(s_k)) \rightarrow P(D_k | s_k)$$

$$f_m(s_k) \rightarrow P(D_k | N)$$

$$P(C|D) = P(S|T=f)$$

$$P(N|D) = P(S|T=f)$$

old s_{total}

$$\lambda_{org}(s) = \frac{P(C|P)}{P(N|D)} = \frac{1}{n} \sum_k \frac{P(D_k | s_k)}{P(D_k | N)} = \frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)} = \frac{1}{n} \lambda_{org}$$

$$\lambda_{org}(s) = \frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)}$$



Old-New =

$$\lambda_{New} = \frac{\lambda_{org}}{\pi \lambda_{org} + (1-\pi)}$$

prob. confusing.

$$\lambda_{org} = \frac{P(C|D)}{P(N|D)} = \frac{P(S|T=f)}{P(S|T=f)} = \frac{\frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)}}{\frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)}} = \frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)}$$

$$\lambda_{new} = \dots = \frac{1}{n} \sum_k \left[\frac{f_m(s_k)}{f_n(s_k)} \right] = \dots = \frac{1}{n} \sum_k \frac{f_m}{f_n}$$

$$\frac{\pi \left[\frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)} \right] + (1-\pi)}{\pi \left[\frac{1}{n} \sum_k \frac{f_m(s_k)}{f_n(s_k)} \right] + (1-\pi)} = \dots$$

$15 + 45 = 60$ $1:3$
 not match with some
 cum e T.

→ Confusing fool:
 ① → (out of all new)

$3/15$ 1st list

$3/12 = \frac{1}{4}$ (25%) 2nd - with list
 $\Rightarrow 1:4$

② out of 30+30/list
 each appearance.

$3/20$
 $4/20 = 1/5$ (20%)

$\therefore$ if 60 last list

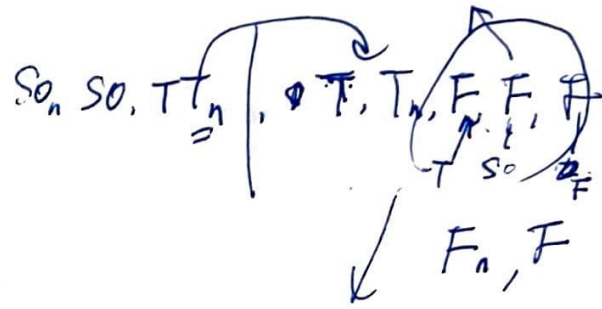
$\Rightarrow 4 \text{ units} \times 3 \text{ tr/unit} = 12 \text{ Traces.}$

$12 \div 3 = 4$

$1:4$
 20% 80%

match
 some current trace
 lost

$12 \times 3 = 36$
 25%
 75



match.
 out of

3 trace.

but actually:

$F(T) \rightarrow 1$

$F(S0) \rightarrow 1$

$F(CF) \rightarrow 1$

4 traces.

$15 = 15$
 25% 75%

$P(\text{activation}) = 3$
 prior

Rich $\Rightarrow$ prior all (if 2n)

$\Rightarrow 2n \times 2 \times \frac{1}{4} = 0.5 \times 2$

$2n \times 2 \times \frac{3}{4} = 1.5 \times 2$

Actually:

$\Rightarrow$ hard to use n

③

∴ prob. match =
(seen only)

$$\frac{\sum 0.5n}{\text{not match?}}$$

→ #activated

$$\frac{\sum 1.5n}{n + \sum 2n} \rightarrow \text{match?}$$

not match & actually no match

total activated trace #

Reich: match

Law of TP:
for a trace: $T=t$
if $T=t_1$
or
 $T=t_2$
∴ $T=t_n$

$$\therefore f(s|T=t_i)$$

$$= f(s|t_1 \cup t_2 \cup \dots \cup t_n)$$

$$= \sum f(s|t_i) \times P(T=t_i|T)$$

i.e. note T if only
T is target present

$$f(s|T) = \sum f(s|t_i) \cdot P(T=t_i|T)$$

$$\frac{P(s|t_i)}{P(s)} = \frac{P(f_n(s_i)) \cdot P(f_n(s_i))}{P(s)}$$

match?

↑

not match?

→ #activated

match?

not match

& actually no match

total activated trace #

Reich: match

i.e. T

$\frac{P(A|B_i)}{P(A|B_j)}$

$= \frac{1}{n} \sum P(A|B_i)$

$P(A)$

$P(A|B_i)$

$P(A|B_j)$

$P(A)$

$P(A|B_i)$

$$\frac{P(s|T=t_k)}{P(s|T=f)} = \frac{P(s|T=t_k) \cdot P(T=t_k|N)}{P(s|T=f) \cdot P(T=f|N)}$$

vs $P(s|T=f)$

× $\frac{1}{n}$ (for $P(s=T)$ in general)

$$\sum_i P(A|B_i) = P(A|B)$$

$A = \{A_1, A_2, \dots, A_n\}$

$$\Rightarrow P(A|B_i) = \frac{P(B_i|A)P(A)}{P(B_i)}$$

$$\Rightarrow P(B_i|A)$$

$$= \frac{P(B_i|A_k) \cdot P(A_k)}{P(A_k)}$$

$$\Rightarrow P(A|B), \text{ when } B = B_1 \cup B_2 \cup \dots \cup B_n$$

$$P(A|B_1 \cup B_2 \cup \dots \cup B_n)$$

$$= \frac{P(A|B_1) + P(A|B_2) + \dots + P(A|B_n)}{P(A|B_1) + P(A|B_2) + \dots + P(A|B_n)}$$

$$P(Ob) \rightarrow P(S|T=t)$$

$$P(Nb) \rightarrow P(S|T=f_{all})$$

$$\frac{P(S|T=t)}{P(S|T=f)}$$

$$T=t = \{T=t_1, T=t_2, \dots, T=t_n\}$$

$$T=f = \{T=f_{all}\}$$

$$\frac{\text{law of total prob.} \sum_i^N P(S|T=t_i) P(T=t_i|T=t)}{P(S|T=f_{all})}$$

$$\frac{P(T=t_i|T=t)}{\frac{1}{n}} \cdot \frac{\sum_i^N P(S|T=t_i)}{P(S|T=f_{all})}$$

$$P(S|T=t_i) = f_{S|T=t_i}(S_1, S_2, \dots, S_n)$$

$$= f_{S_1, \dots, S_n}(S_1, S_2, \dots, S_n | T=t_i)$$

$$= f_{S_1}(S_1 | T=t_i) \times \prod_{i \neq 1} f_{S_i}(S_i | T=t_i)$$

$$= f_m(S_k) \prod_{i \neq k} f_n(S_i)$$

(? Similarity $A \sim S$
& T/F RV
independence?

$$S \rightarrow S(w_1, w_2)$$

$$T \rightarrow T(w_1) \dots ?$$

MDS: $p_1, p_2, p_3, \dots, p_{dim}$

Word 1	E_0	0	0	...	0
Word 2	0	0	0	...	0

REM. D_x wird.

$$[P_{val}, \dots, D_{val}]$$

$$f_m(S_k) = P(D_k = S_k | S_k)$$

REN paper

here $f_{S_k}(S_k | T=t_i) ?$

$$\Rightarrow f_m(S_k)$$

$$= P(D_k = S_k | S_k)$$

$$= f_{S_k}(S_k = S_k | T=t_i)$$

$$f_n(S_k) \rightarrow P(D_i | N_i) \text{ i.e. } P(D_i | N_i)$$

$$f_{S_k}(S_k = S_k | T=t_k)$$

$i \neq k$

$$\frac{1}{n} \sum_k [f_m(S_k) \prod_{i \neq k} f_n(S_i)]$$

$$= \frac{1}{n} \sum_k \frac{f_m(S_k)}{f_n(S_k)} \cdot \frac{1}{n} \sum_{i \neq k} f_n(S_i)$$

decide by $f_n(S_i)$

$$= \frac{1}{n} \sum_k \frac{P(D_k | S_k)}{P(D_k | N_k)}$$

$$= \frac{1}{n} \sum_k \lambda_k$$

$$= f_m(S_k) \prod_{i \neq k} f_n(S_i)$$